

Sign Reversal of the Capacitive Memory Window in $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ Ferroelectric Transistors: Endurance Failure Without Polarization Loss

Apu Das, Agniva Paul, Debraj Barman, Zhao-Feng Lou, Yannick Raffel, Min-Hung Lee, Sourav De,

Abstract—Endurance failure in HfO_2 ferroelectric transistors is routinely attributed to ferroelectric fatigue. We show that it is not. In a $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{SiO}_2(0.85\text{ nm})/\text{p-Si}$ transistor, the DC-sweep C - V hysteresis window falls from 0.41 V to zero at 10^3 cycles and reverses sign to -0.10 V at 10^4 , on both the flatband and threshold flanks. Over the same range the remanent polarization increases: $2P_r$ rises 45% to $29.2\text{ }\mu\text{C cm}^{-2}$ by 10^2 cycles and remains above its pristine value at 10^4 , with sharp switching peaks and coercive voltage constant to 4%. Series-resistance-corrected accumulation capacitance is unchanged to within 2%, excluding any electrostatic change in the stack. The window efficiency consequently falls from 8.3% to -1.5% . Polarization degrades only at 10^5 cycles, two decades after the device has ceased to store information. Interfacial charge first compensates, then over-compensates, the polarization field at the channel.

Index Terms—Ferroelectric field-effect transistor, hafnium oxide, endurance, memory window, charge trapping, capacitance-voltage, interfacial layer, PUND.

I. INTRODUCTION

FERROELECTRICITY in doped HfO_2 [1], [2] made a CMOS-compatible ferroelectric field-effect transistor (FeFET) practical, and integrated FeFET technologies now exist at the 28 nm node and below [3], [4]. Endurance remains the principal obstacle: the memory window of an HfO_2 FeFET typically collapses between 10^4 and 10^5 write cycles [5], [6], orders of magnitude short of a non-volatile memory requirement, and the gap persists across silicon-doped and zirconium-doped stacks alike [7], [8].

The default explanation is ferroelectric fatigue — progressive loss of switchable polarization through domain-wall pinning and defect redistribution [9]. An alternative attributes

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Apu Das, Agniva Paul, Debraj Barman, and Sourav De are with the College of Semiconductor Research, National Tsing Hua University, Hsinchu 30044, Taiwan (e-mail: apu.das@gapp.nthu.edu.tw; sourav.de@mx.nthu.edu.tw).

Zhao-Feng Lou and Min-Hung Lee are with National Taiwan University, Taipei 10617, Taiwan. Yannick Raffel is with the Center Nanoelectronic Technologies, Fraunhofer Institute for Photonic Microsystems IPMS, Dresden 01109, Germany.

Corresponding authors: Apu Das and Sourav De.

the collapse to charge injected into the ferroelectric/interfacial-layer region, which screens the polarization from the channel without destroying it [10], [11]. The two mechanisms imply opposite remedies — ferroelectric materials engineering in the first case [12], [13], interface engineering in the second [14], [15] — so distinguishing them matters. Reported evidence supports both: oxygen-vacancy redistribution during cycling is well documented [16]–[18], as is the growth of interfacial trap density measured by charge pumping [19], low-frequency noise [20], and trap capture–emission spectroscopy [21], [22].

We have previously shown on this stack family that polarization survives the collapse of the memory window [23], and that pulse shaping extends endurance in naturally aged devices [24]. A companion preprint reports the stability of the gate-stack polarization response in degraded devices [?]. What has not been established is how far the decoupling extends, whether it can be resolved capacitively on the transistor itself, and what happens to the window once interfacial compensation exceeds the polarization contribution.

Here we report a qualitatively stronger result. Across the cycling range in which the device stops functioning as a memory, the remanent polarization does not merely survive — it *exceeds* its pristine value. And the window does not merely vanish: it changes sign. Beyond $\sim 10^3$ cycles a positive gate bias leaves the transistor harder to turn on, the opposite of its pristine behavior. We show this occurs at constant gate-stack capacitance, and that ferroelectric degradation appears only at 10^5 cycles, two decades later.

II. DEVICE FABRICATION AND MEASUREMENT

A. Gate Stack

The device is a p-Si channel FeFET with a $\text{TaN}/\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2(10\text{ nm})/\text{SiO}_2(0.85\text{ nm})$ gate stack, shown in Fig. 1(a). The chemical SiO_2 interfacial layer (IL) is unavoidable on a silicon channel and is the region of interest in what follows. The insulator stack forms a series capacitive divider,

$$\frac{V_{\text{HZO}}}{V_{\text{ins}}} = \frac{C_{\text{IL}}}{C_{\text{IL}} + C_{\text{HZO}}}, \quad (1)$$

where $V_{\text{ins}} = V_{\text{HZO}} + V_{\text{IL}}$. With $\varepsilon_{\text{HZO}} = 30$ and $\varepsilon_{\text{IL}} = 3.9$, $C_{\text{HZO}} = 2.66\text{ }\mu\text{F cm}^{-2}$ and $C_{\text{IL}} = 4.06\text{ }\mu\text{F cm}^{-2}$, giving a ratio of 0.60. Because the semiconductor capacitance $C_S(V_G)$ is in series and is neglected in (1), 0.60 is an upper bound on the fraction of gate bias reaching the ferroelectric. The measured accumulation capacitance of 196 pF agrees with this stack to within 8% for a $100\text{ }\mu\text{m} \times 100\text{ }\mu\text{m}$ gate.

B. Polarization Measurement

Switching-current and P - E measurements used triangular waveforms with a half-period of 500 μ s. Polarization was evaluated by positive-up-negative-down (PUND) measurement. The positive and negative switching charges were determined from

$$Q_{\text{sw}}^+ = \int [I_P(t) - I_U(t)] dt, \quad Q_{\text{sw}}^- = \int [I_N(t) - I_D(t)] dt, \quad (2)$$

where P and N denote the switching pulses and U and D the corresponding nonswitching pulses. The switchable polarization was then calculated by averaging the magnitudes of the positive and negative switching charges,

$$2P_r = \frac{|Q_{\text{sw}}^+| + |Q_{\text{sw}}^-|}{2A}. \quad (3)$$

The positive and negative coercive voltages, V_c^+ and V_c^- , were obtained from the zero crossings of the P - V branches, and the imprint voltage and coercive field from

$$V_{\text{imp}} = \frac{V_c^+ + V_c^-}{2}, \quad E_c = \frac{|V_c^+| + |V_c^-|}{2d}. \quad (4)$$

Applying (4) to the pristine device gives $V_c = 2.08$ V; combined with the 0.60 divider of (1) this corresponds to $E_c \approx 1.25$ MV/cm in the ferroelectric, consistent with reported values for $\text{Hf}_{0.5}\text{Zr}_{0.5}\text{O}_2$ [25], [26]. This measured E_c is used below rather than a literature value.

C. Endurance Cycling and Capacitance Measurement

Endurance cycling used ± 5 V, 50 μ s pulses. At each checkpoint ($N = 0, 1, 10, 10^2, 10^3, 10^4, 10^5$) the device was characterized by PUND and by C - V on a Keysight B1500A CMU in C_p - R_p mode with a 20 mV AC level, at six frequencies (10 kHz–1 MHz) and four sweep amplitudes (± 1.5 V– ± 3 V), with source, drain and body tied so that the inversion branch is supplied with minority carriers [27]. The protocol is summarized in Fig. 1(b). In total 290 C - V sweeps were acquired.

Two features of the capacitance measurement require explicit statement. The sweep begins at the negative rail and integrates for one power-line cycle per point, so a full sweep takes ~ 1 s — four orders of magnitude slower than the write pulse. Consequently the pulsed state does not survive to mid-sweep, and C - V curves acquired after program and after erase are indistinguishable (Fig. 3). The quantity reported here is therefore the *DC-sweep hysteresis window*

$$\Delta V_{\text{hyst}} = V_{\text{fwd}} - V_{\text{rev}} \Big|_{C=C_{\text{min}} + \frac{1}{2}(C_{\text{max}} - C_{\text{min}})}, \quad (5)$$

the separation of the forward and reverse branches at 50% of the C - V flank. It is not the retained state of a pulse-written bit. It nevertheless probes the same polarization-to-channel coupling that sets ΔV_{th} , since both are determined by how much polarization charge reaches the inversion layer rather than being compensated at the interface. The sign convention is fixed throughout: $\Delta V_{\text{hyst}} > 0$ denotes a counter-clockwise loop, in which a positive sweep leaves V_{th} lower (the ferroelectric sense); $\Delta V_{\text{hyst}} < 0$ denotes a clockwise loop, in which a positive sweep leaves V_{th} higher (the electron-trapping sense).

D. Series-Resistance Correction and Screening

Because R_p is measured directly, the parallel conductance $G = 1/R_p$ is available. A series resistance was extracted at each checkpoint from the Nicollian-Brews strong-accumulation expression [?], [?]

$$R_s = \frac{G_{ma}}{G_{ma}^2 + \omega^2 C_{ma}^2}, \quad (6)$$

evaluated at the highest valid frequency, where G_{ma} and C_{ma} are the measured accumulation values. Capacitances were then corrected using

$$C_c = \frac{(G^2 + \omega^2 C^2) C}{a^2 + \omega^2 C^2}, \quad a = G - (G^2 + \omega^2 C^2) R_s. \quad (7)$$

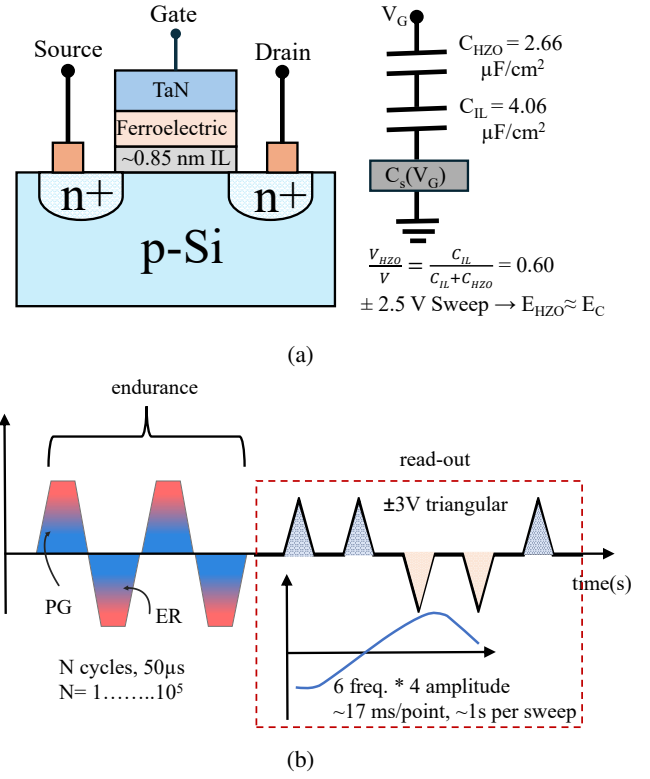


Fig. 1: Device and measurement protocol. (a) Gate stack and equivalent series divider, (1). With $C_{\text{IL}} = 4.06 \mu\text{F cm}^{-2}$ and $C_{\text{HZO}} = 2.66 \mu\text{F cm}^{-2}$, at most 60% of the gate bias reaches the ferroelectric; with the measured $V_c = 2.08$ V this gives $E_c \approx 1.25$ MV/cm. (b) At each checkpoint the device is cycled with ± 5 V, 50 μ s pulses, then characterized by PUND and C - V . The C - V sweep starts at the negative rail and takes ~ 17 ms per point, four orders of magnitude slower than the write pulse; the reported window is therefore DC-sweep hysteresis, not a pulse-written state.

Sweeps with a median loss tangent $D = 1/\omega R_p C_p \geq 0.9$, for which the parallel model is not meaningful, were excluded from window statistics; this removed 57 of 290 sweeps, of which 38 are at $N = 10^4$. The correction of (7) is below 5% through 10^2 cycles, so results in that range are essentially uncorrected.

E. Figure of Merit

An unscreened remanent polarization would produce a window $2P_r/C_{IL}$. We therefore define the window efficiency

$$\eta = \frac{\Delta V_{\text{hyst}}}{2P_r/C_{IL}} \quad (8)$$

as the fraction of the polarization that reaches the channel. Where a window deficit is attributed to interfacial charge, the equivalent areal density follows from

$$\Delta N_{\text{it}} = \frac{C_{IL} \Delta V}{q} \approx \Delta V \times 2.5 \times 10^{13} \text{ cm}^{-2} \text{ V}^{-1}. \quad (9)$$

III. RESULTS & DISCUSSIONS

A. The Ferroelectric Remains Intact to 10^4 Cycles

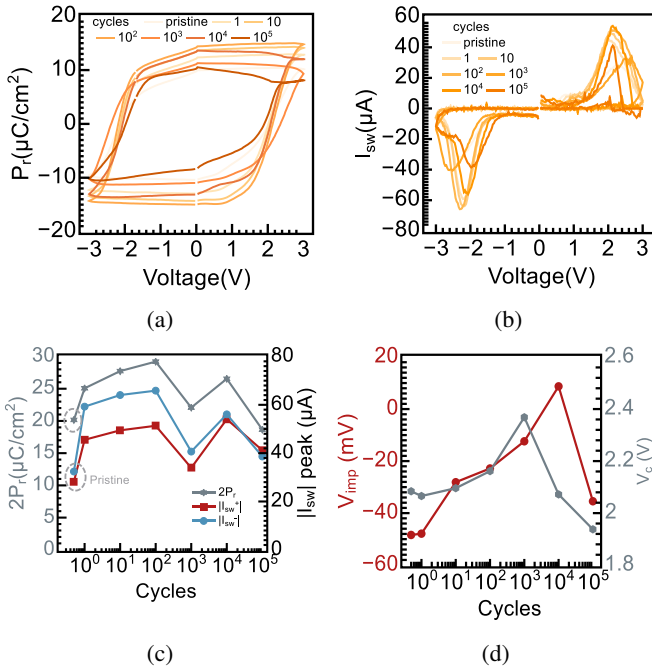


Fig. 2: The ferroelectric remains intact to 10^4 cycles. (a) P - V loops and (b) switching current, 500 μs half-period triangular waveform. (c) $2P_r$ from (3) (left) and switching peaks (right); dotted line marks the pristine $2P_r$. Wake-up raises $2P_r$ by 45% by 10^2 cycles and it stays above pristine through 10^4 . (d) V_{imp} and V_c from (4). V_c is constant to 4% through 10^4 , so the switching threshold does not move while the window is lost; drifts monotonically from -48 mV to 9 mV over the same range, crossing zero between 10^3 and 10^4 . At 10^5 , V_c decreases as the peaks broaden — not a domain-pinning signature. Open symbol at 10^3 : single unverified trace.

Fig. 2 shows the polarization and switching-current response. Field cycling produces a pronounced wake-up, consistent with field-driven oxygen-vacancy redistribution and domain-wall de-pinning reported for this material system [17], [28], [29]: $2P_r$ from (3) rises from $20.2 \mu\text{C}/\text{cm}^2$ pristine to $29.2 \mu\text{C}/\text{cm}^2$ at 10^2 cycles, a 45% increase, and the positive switching peak nearly doubles from $28 \mu\text{A}$ to $51 \mu\text{A}$. Both remain at or above their pristine values through 10^4 cycles.

The switching peaks stay sharp and V_c from (4) is constant to within 4% at every verified checkpoint through 10^4 (Table II), so neither the nucleation-limited character of the switching nor the coercive field has changed.

Degradation appears only at 10^5 cycles, where $2P_r$ falls to $18.6 \mu\text{C}/\text{cm}^2$, below the pristine value, the peaks broaden and drop, and the non-switching baseline current rises. Notably V_c decreases at this point, from 2.08 V pristine to 1.94 V, which is not the signature of domain-wall pinning; together with the non-switching current I_{ns} reaching its highest value (Table I) it is more consistent with a leakier, electrically softened film, in which trap-assisted Poole-Frenkel conduction becomes significant [30]–[32]. The distinction matters for the argument that follows: the ferroelectric does eventually fatigue, but it does so two decades after the device has ceased to function.

The imprint voltage of (4) drifts monotonically from -48 mV pristine to 9 mV at 10^4 cycles, crossing zero between 10^3 and 10^4 [Fig. 2(d)], consistent with the gradual accumulation of a negative charge sheet on the channel side of the ferroelectric; a cycling-induced internal bias field of this kind is well established in both perovskite and hafnia ferroelectrics [26], [33], [34]. At 10^5 cycles it returns to -35 mV, coincident with the onset of polarization loss; we therefore treat the imprint trend as supporting evidence over $N \leq 10^4$ only and draw no conclusion from the 10^5 point.

B. The Window Collapses and Reverses Sign

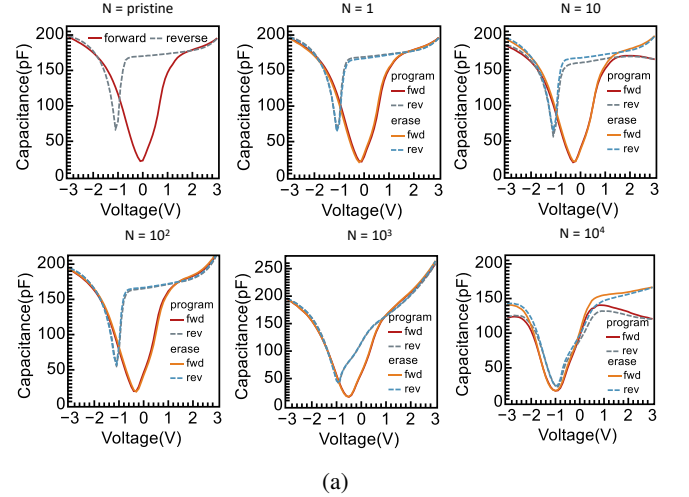


Fig. 3: C - V families at 100 kHz, ± 3 V, corrected by (7). Solid: forward; dashed: reverse. Red and blue are post-program and post-erase; their coincidence shows the pulsed state does not survive the sweep, hence the use of (5). The forward-reverse separation narrows monotonically, closes near 10^3 cycles, and reopens at 10^4 with the branches exchanged. Raw data shown at 10^4 , where the lumped C_p - R_p model fails.

Fig. 3 shows the C - V families. The curves are MOS-like with a full inversion branch. Curves measured after program and after erase coincide at every checkpoint, confirming that

the pulsed state does not survive the sweep. The separation between forward and reverse branches narrows monotonically with cycling, closes near 10^3 , and reopens at 10^4 with the branches exchanged.

Fig. 4 quantifies this. ΔV_{hyst} from (5) falls from $+0.41$ V pristine to $+0.03 \pm 0.02$ V at 10^3 — statistically indistinguishable from zero — and to -0.10 ± 0.02 V at 10^4 , negative by 4.6σ . The behavior reproduces at all four sweep amplitudes.

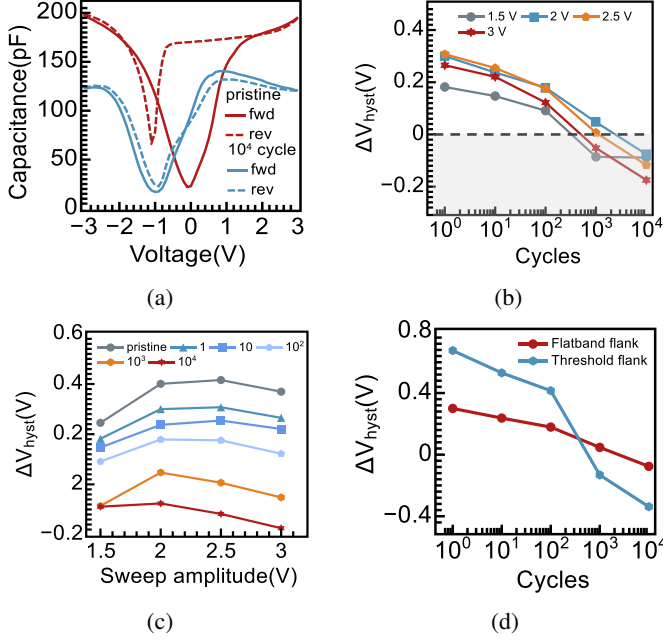


Fig. 4: The window collapses and reverses sign. (a) C - V at 100 kHz, pristine and 10^4 cycles. (b) ΔV_{hyst} versus cycling at four amplitudes; shaded region marks clockwise (trapping-sense) loops. (c) ΔV_{hyst} versus sweep amplitude, peaking at ± 2.5 V where (1) places E_{HZO} just above the measured E_c — switching completion. (d) Flatband versus threshold flank at ± 2 V. Both invert on the same schedule, excluding an extraction artifact; the 2.1 – $2.3\times$ ratio indicates loop stretching, hence bias-dependent interface charge.

Three further observations constrain the interpretation. First, the window peaks at a ± 2.5 V sweep and decreases at ± 3 V. From (1) a 2.5 V sweep places E_{HZO} at 1.5 MV/cm, just above the measured $E_c = 1.25$ MV/cm; the maximum therefore marks switching completion, and the amplitude axis is effectively a write-voltage axis. The agreement between an amplitude identified purely from the C - V response and a coercive field measured independently by PUND is a consistency check on the divider of (1) and on the nucleation-limited switching picture [25].

Second, the window measured on the inversion flank — the flank corresponding to V_{th} — collapses and inverts on the same schedule, excluding an artifact of the extraction point. That window is consistently 2.1 – $2.3\times$ the flatband-flank window (Table I), indicating that the loop stretches rather than translating rigidly. A rigid shift, as produced by a fixed charge sheet, would displace both flanks equally; a flank-dependent

shift requires charge that fills and empties as the surface potential is swept, i.e. bias-dependent interface states.

Third, ΔV_{hyst} is suppressed at low frequency and saturates above ~ 100 kHz following

$$\Delta V_{\text{hyst}}(f) = \Delta V_{\infty} - \frac{\delta V}{1 + (f/f_c)^2}, \quad (10)$$

with $f_c = 26$ kHz– 34 kHz ($\tau = 4.7$ μ s– 6.0 μ s) that does not shift with cycling. Charge that is given time to respond screens the window; charge that cannot follow the AC signal does not. That f_c is cycle-invariant while δV is not shows that cycling changes how much charge participates, not the nature of the process.

C. Polarization and Window Are Decoupled

Fig. 5 places the two measurements together and constitutes the central result. Between the pristine state and 10^2 cycles the polarization *increases* by 45% while the window it delivers to the channel *falls* by 57%. The window efficiency of (8) declines monotonically from 8.3% to 0.5% over the same range and becomes negative at 10^4 (Table ??), where the ferroelectric holds 32% more remanent polarization than pristine and the channel nevertheless sees a window of the opposite sign. This inverts the conventional reading of memory-window closure, in which the window is assumed to track the switchable polarization [35], [36].

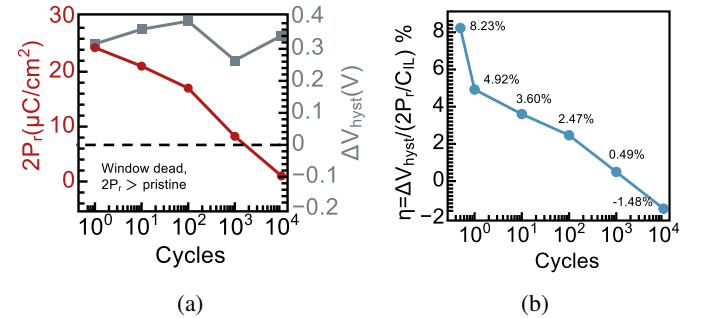


Fig. 5: Polarization and window are decoupled. (a) $2P_r$ (left) and ΔV_{hyst} (right) versus cycling. Between 1 and 10^2 cycles the polarization rises 45% while the window falls 41%. In the shaded region the window is dead or inverted while $2P_r$ exceeds pristine (dotted line). (b) Window efficiency (8), falling from 8.3% to -1.5% . Error bars: standard deviation over two amplitudes and two frequencies. Open symbol at 10^3 : single unverified P - V trace..

D. The Gate Stack Is Unchanged

The obvious alternative — that the stack itself degraded — is excluded by Fig. 6(a). The raw accumulation capacitance appears to fall by 24% by 10^3 cycles, but this is entirely a series-resistance effect: after correction by (7), $C_{\text{acc}} = 192.0 \pm 3.7$ pF across all nine program- and erase-state measurements from the pristine device through 10^3 cycles, spanning 183 pF–196 pF. The magnitude of the correction is itself informative: it is below 5.5% at every checkpoint through 10^2 cycles and reaches 26%–31% only at 10^3 , so the constancy of C_{acc} over

the range where the window loses most of its magnitude rests on essentially uncorrected data. There is no interfacial regrowth, no thinning, and no change in how the gate couples to the channel. At 10^4 cycles the lumped C_p – R_p model fails and no corrected value is quoted. No pristine measurement in the program state was acquired.

The loss spectra of Fig. 6(b) separate two dissipation channels by their frequency signature. The high-frequency branch, $D \propto f$, is series resistance and is present at all checkpoints. The low-frequency branch, $D \propto 1/f$, is conduction; it is negligible pristine and dominant from 10^3 onward. Fig. 6(c) shows the accumulation conductance rising from $3.3 \mu\text{S}$ at 10^2 cycles to $115 \mu\text{S}$ at 10^3 , a factor of 35, with program and erase states agreeing to within 2% — the same decade in which the window changes sign.

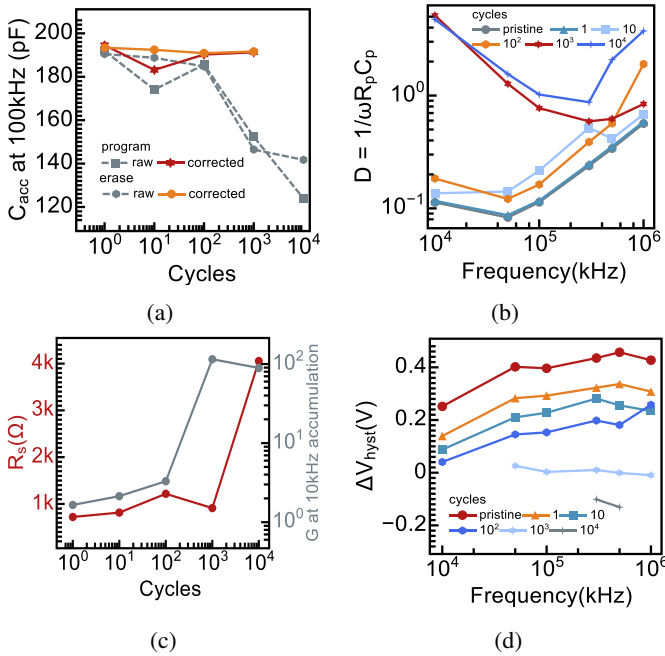


Fig. 6: The stack is unchanged; a conduction path opens. (a) C_{acc} at 100 kHz, raw (dotted) and corrected by (7) (solid), against the nominal stack value (dashed). The apparent 24% raw decrease is entirely a series-resistance effect; corrected, $C_{\text{acc}} = 192.0 \pm 3.7 \text{ pF}$ from pristine through 10^3 , with the correction below 5.5% through 10^2 cycles. Shading: mean ± 1 s.d. No pristine program-state measurement was acquired. (b) Loss spectra: the $D \propto f$ branch is series resistance, the $D \propto 1/f$ branch is conduction, absent pristine and dominant from 10^3 . Dotted line: screening threshold $D = 0.9$. (c) R_s from (6) (left) and accumulation conductance (right), which rises 35-fold between 10^2 and 10^3 cycles — the decade in which the window changes sign. (d) ΔV_{hyst} versus frequency with fits to (10); $f_c = 26$ – 34 kHz does not shift systematically with cycling.

E. Interpretation

Three facts hold simultaneously at 10^4 cycles: the polarization is present and larger than pristine; the gate-stack capacitance is unchanged; and the window the channel experiences

has reversed sign. Only one class of explanation survives. Charge must be accumulating *between* the ferroelectric and the silicon — nothing else can screen the polarization from the channel while leaving both the polarization and the total series capacitance intact. From (9), the 0.51 V swing in ΔV_{hyst} between the pristine and 10^4 states corresponds to $\Delta N_{\text{it}} \approx 1.3 \times 10^{13} \text{ cm}^{-2}$, a density readily accommodated at an $\text{HfO}_2/\text{SiO}_2/\text{Si}$ interface [22], and comparable to the trap densities extracted by charge pumping and low-frequency noise on nominally identical stacks [19], [20].

TABLE I
POLARIZATION, MEMORY WINDOW AND WINDOW EFFICIENCY

N	$2P_r$ ($\mu\text{C cm}^{-2}$)	$2P_r/C_{\text{IL}}$ (V)	ΔV_{hyst} (V)	$\Delta V_{\text{hyst}}^{\text{inv}}$ (V)	η (%)
pristine	20.2	4.96	$+0.41 \pm 0.03$	$+0.83$	$+8.2$
1	25.1	6.17	$+0.30 \pm 0.03$	$+0.68$	$+4.9$
10	27.7	6.83	$+0.25 \pm 0.03$	$+0.53$	$+3.6$
10^2	29.2	7.19	$+0.18 \pm 0.02$	$+0.42$	$+2.5$
10^3	22.1	5.43	$+0.03 \pm 0.02$	-0.13	$+0.5$
10^4	26.5	6.53	-0.10 ± 0.02	-0.34	-1.5
10^5	18.6	4.59	—	—	—

TABLE II
FERROELECTRIC SWITCHING DIAGNOSTICS

N	V_c (V)	E_c (MV cm^{-1})	V_{imp} (mV)	$I_{\text{sw}+}$ (μA)	$I_{\text{sw}-}$ (μA)	I_{ns} (μA)
pristine	2.08	1.25	-48	28.2	-32.4	3.15
1	2.07	1.24	-48	45.5	-59.3	1.46
10	2.10	1.26	-28	49.4	-64.1	0.83
10^2	2.16	1.30	-23	51.3	-65.9	0.62
10^3	2.37	1.42	-12	33.9	-40.7	1.52
10^4	2.07	1.24	+9	54.0	-56.1	1.20
10^5	1.94	1.16	-35	41.2	-38.6	1.87

The sign reversal further shows that this charge does not merely neutralize the polarization's influence. It exceeds it. Past $\sim 10^3$ cycles the threshold voltage follows the trapped charge rather than the ferroelectric, so the device does not simply lose its memory: it inverts it. A memory returning the wrong state on read is a more severe failure mode than one with no window at all, and there is necessarily an intermediate cycle count at which the two contributions cancel exactly. The three independent signatures — the flank asymmetry, the imprint drift of Fig. 2(d), and the conduction onset of Fig. 6(c) — all point to the same interfacial region and all develop over the same cycling range.

This extends our earlier finding that polarization survives window collapse [23] in two respects. The polarization is not merely preserved but enhanced across the failure range, removing any residual possibility that a small undetected polarization loss accounts for the collapse. And the failure is resolved capacitively, without pulsed characterization, on the transistor itself. That a device can retain — indeed improve — its polarization while ceasing to function as a memory also bears on reliability projections for in-memory and neuromorphic

applications, where FeFETs are operated in multi-level rather than binary states [37].

Two points of caution should be stated. The $\tau = 4.7\mu\text{s}$ – $6.0\mu\text{s}$ population probed by the AC dispersion of (10) is not the population responsible for the DC sign reversal, which is established over the $\sim 17\text{ms}$ dwell of the sweep; these are distinct states separated by more than three decades in time constant, and we do not claim a common origin. Separately, δV in (10) decreases with cycling rather than growing, so the AC-active interfacial population is not itself the one that inverts the loop.

Three limitations follow. The results are from a single device; cycling is destructive, so checkpoints cannot be repeated on the same transistor. The value of η depends on the convention chosen for the ideal window in (8); alternative denominators shift the absolute values but not the sign change or the trend. Finally, the spatial location of the compensating charge is inferred by elimination rather than imaged; transmission electron microscopy with electron energy-loss spectroscopy would localize it directly.

IV. CONCLUSION

The memory window of an HfO_2 FeFET does not fail because the ferroelectric fails. In the device studied here the DC-sweep hysteresis window collapses to zero at 10^3 cycles and reverses sign by 10^4 , while $2P_r$ stands 32% above its pristine value and the gate-stack capacitance is unchanged. Ferroelectric degradation arrives only at 10^5 cycles, two decades after the device has stopped storing information. Endurance in this technology is therefore limited by the electrostatic coupling between the polarization and the channel, not by the polarization itself, and interface engineering rather than ferroelectric engineering is the productive direction.

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